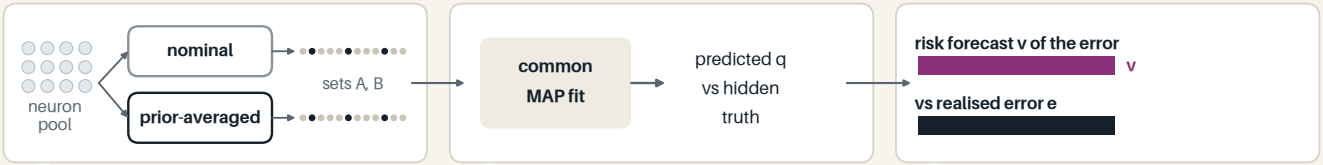
Refined risk estimates often leave the recording decision unchanged

Choosing 3 neurons to record in order to predict a stimulus response; simulated 12-neuron networks with known wiring



1 Same recordings?

Do the criteria choose different data? (40 networks)

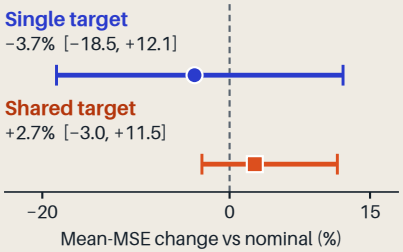
68% 82 of 120 single-target designs are identical

65% 26 of 40 shared-target designs are identical

same recording set

2 Lower response error?

Actual MSE after the common fit (40 networks)

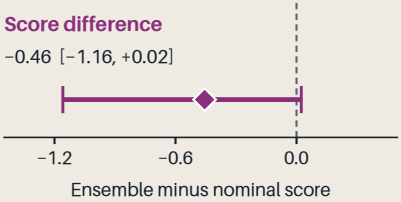


97.5% paired network-cluster CI

3 Better forecast?

Same data and prediction (40 new networks)

19 of 40 networks better, 21 worse



One network supplies 64% of the summed difference; 95% paired CI

WHAT THIS SHOWS

How much prior averaging can help is capped by how often it changes the design. All three primary intervals include zero, so no stable gain is shown. The three-layer audit can be applied to other criteria.

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Simulation only; no biological validation